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FHRDiff: Leveraging Diffusion Models for Conditional Fetal Heart Rate Signal Generation

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***Abstract*—Accurate analysis of Fetal Heart Rate (FHR) signal is often impeded by challenges such as data scarcity and label imbalance, which affect the reliability and robustness of deep learning models. To address these challenges, this study introduces FHRDiff, a novel diffusion model conditioned on Phase-Rectified Signal Averaging (PRSA) spectrograms for the creation of synthetic FHR signals. Our model integrates time encoding, condition generation from PRSA spectrograms, and residual blocks with dilated convolutions to effectively manage temporal dynamics and long-range dependencies. Extensive qualitative and quantitative experiments on FHR signal synthesis demonstrate the feasibility and effectiveness of FHRDiff. Compared to Generative Adversarial Networks (GANs) and image-based diffusion model, our method achieves the highest signal fidelity and distribution similarity, with key measures including 0.067 maximum mean deviation (MMD), 0.492 percent root mean square difference (PRD), 1.763 relative entropy (RE), and 0.160 Frechet distance (FD). Expert validation confirms the model's capacity to accurately generate data for normal and abnormal FHR signals based on the paired condition. In addition, an ablation study was conducted to highlight that the spectrogram paired condition can guide the diffusion model to produce synthetics FHR signals with greater diversity compared to unconditional models. The results emphasize diffusion models' potential for broad application in biomedical time series analysis, such as generation, imputation and noise removal.**

***Index Terms*—Diffusion Models, PRSA, FHR, Cardiotocography**

I. Introduction

Over the last century, stillbirth rates have decreased by 90% due to remarkable advancements in obstetric care technologies [1], [2]. Timely monitoring and detection of risks for both the fetuses and pregnant women are essential for preventing major pregnancy complications and are of great importance in public health. Among these techniques, cardiotocography (CTG) has been the standard method for monitoring fetal well-being [3], [4].

Fetal Heart Rate (FHR) monitoring is a crucial diagnostic technique that can promptly identify fetuses at risk and determine the optimal timing for delivery. However, the accuracy and effectiveness of manual diagnosis for FHR data heavily depend on the clinician's experience, leading to significant subjective variability both between different observers (inter-observer) and within the same observer over time (intra-observer). These challenges are especially pronounced with FHR data, which are characterized by long sequences (may up to 90 minutes), rapid variability, complexity, and a high susceptibility to numerous artifacts. Therefore, numerous advanced Computer-Aided Diagnosis (CAD) methods based on AI and deep learning have been developed to address these issues [5]–[8].

Enhancing the performance of deep learning models in FHR research necessitates a large amount of clinically labeled data. However, obtaining FHR data is challenging due to three main issues: data scarcity, class imbalance, and concerns about patient privacy. As far as we know, there are few available FHR dataset such as CTU-UHB with 552 recordings [9], [10] and CULF-DB with only 155 recordings [11], [12]. Additionally, normal patterns are substantially more common than pathological ones, maintaining a 4:1 ratio in CTU-UHB dataset, which makes the classification task harder. Therefore, developing a generative model to synthesize high-quality FHR recordings can effectively address these issues.

In this study, we aim to address a fundamental scientific question: *How can advanced generative models be applied to produce high-quality synthetic FHR signals that accurately reflect the complex characteristics of real FHR data?* It encompasses two challenges: (1) **Maintaining Signal Fidelity**: Assessing the ability of diffusion models to preserve signal fidelity and avoid issues in Generative Adversarial Networks (GANs) such as model collapse; (2) **Pattern Differentiation**: Evaluating how effectively these models can distinguish between normal and pathological patterns in FHR data.

Current developments in synthetic FHR signal generation can be classified into traditional methods and deep learning-based methods. Traditional methods generate new samples from existing data or based on specified formulas [13]–[15], but they suffer from limited diversity and variation. Deep learning-based methods, primarily using GANs [16], can produce high-quality biological signals. However, these models, whether unconditional or using classifiers as conditions, also lack diversity in synthetic data.

Recently, diffusion models have shown remarkable abilities

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in generating high-quality images [17]–[19], videos [20]–[22], and time series data [23]–[25] since the introduction of Denoising Diffusion Probabilistic Models (DDPMs) [26]. Unlike GANs, which can suffer from mode collapse, diffusion models avoid this issue and can generate entirely new data with high fidelity, significantly enhancing the diversity of synthetic data.

In this work, we present a novel framework for generating realistic FHR signals using diffusion probabilistic models enhanced by Phase-Rectified Signal Averaging (PRSA) for detailed spectral analysis. Our method employs a two-phase diffusion process and a reverse process. PRSA captures spectral characteristics by identifying deceleration anchor points and transforming them into spectrograms via Continuous Wavelet Transform (CWT). These spectrograms guide the signal generation, capturing both short-term and long-term FHR variations. Our model architecture integrates time encoding, condition generation from PRSA spectrograms, and residual blocks with dilated convolutions to manage temporal dynamics and long-range dependencies.

Therefore, the main contributions of this study are as follows:

- **Formulating CTG signal generation as a generative denoising process.** This is the first study to apply the diffusion model to CTG signal generation, solving the problem of model collapse and generating high-fidelity data with a distribution similar to real FHR data.
- **Introducing the PRSA curve for generating paired spectrogram conditions.** This method enhances the model's ability to distinguish between normal and pathological features in raw FHR data, providing precise and effective spectral guidance during the signal generation process.
- **Conducting a thorough comparison with several state-of-the-art GAN models tailored for FHR signal generation.** This comparison includes various evaluation metrics to rigorously assess the quality and fidelity of the generated data.

II. Related Work

A. Methods for Generating Synthetic FHR Signals

Currently, there are few methods for synthetic FHR signal generation due to several factors, such as complexity of FHR signals, lack of comprehensive datasets, and validation challenges. Most remain at the level of small-sample generation technology, commonly known as data augmentation. Data augmentation involves generating or synthesizing new data from existing data. These methods primarily include oversampling techniques, mathematical-based methods, and intelligent methods.

Oversampling techniques, such as random flipping, random translation, and random cropping, are unsuitable for FHR signals as they disrupt the physiological and pathological characteristics of these signals [13]. Mathematical-based methods generate signals based on specified formulas [14], [15], which results in a lack of diversity. Intelligent methods mainly encompass deep learning-based models, such as GANs and their improved algorithms. However, GANs face the problem of mode collapse, where the model can only learn and generate a partial distribution of the training data.

Existing deep learning approaches to synthesizing FHR signals are all based on GANs. FHRGAN [16] incorporates an auxiliary classifier to generate multi-category FHR signals from noise, distinguishing between normal and abnormal samples. However, its ability to enhance the diversity of generated signals is limited by the use of only two categories and the minor differences in Gaussian noise used as input data. CTG-GAN [27] integrates self-attention and residual modules within a Conditional GAN framework, finely tuned to accurately replicate the complex patterns characteristic of FHR data. Despite effectively capturing overall variability, the model falls short in emulating some highly specific characteristics of FHR signals. There is potential for further enhancement of the model's generalization capabilities, particularly through the utilization of a larger, real-world dataset.

B. Diffusion Models for Generating Biomedical Time Series Signals

Diffusion models are advanced machine learning algorithms that generate high-quality data by progressively adding noise to a dataset and then learning to reverse this process. These models have shown great potential in modeling complex data distributions, particularly in image generation and data synthesis. Inspired by their success in image synthesis, some studies have explored their capability in processing and generating time-series signals. Biodiffusion is a diffusion model tailored for biomedical signal synthesis [28], which was tested on three types of data: simulated signals, accelerometer signals, and ECG signals. Biodiffusion can be used for various tasks, including unconditional, label-conditional, and signal-conditional generation, making it suitable for applications such as noise removal and interpolation.

Additionally, a conditional diffusion model, TimeDiff [29], was proposed to generate complete multivariate time series from incomplete data. Tosato et al. [30] introduced a class-conditioned diffusion model that generates ECG recordings. They first converted the signals into electrode-frequency distribution maps (EFDMs), trained the model on these EFDMs, and then generated new signals. Another diffusion model for synthesizing ECG signals was proposed in [31]. However, this model maps 1D ECG data into 2D space and converts them into image files, which results in worse performance compared to the Wassertein GAN with Gradient Penalty (WGAN-GP) [32], which processes input/output data in 1D space.

Existing diffusion models, whether they are unconditional diffusion models, class-conditioned diffusion models, or models that use incomplete signals as conditions to perform interpolation, limit the diversity of the generated data or lack high fidelity. In this work, we propose a novel diffusion model that uses PRSA curve spectrograms as a condition to generate FHR signals. We use FHR signal-PRSA spectrogram

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pairs as training data, meaning each input FHR signal has a unique label. This label has been proven to capture important physiological and pathological features of the FHR signals [33] and will guide the model, significantly enhancing the diversity of the generated signals. This model is capable of generating FHR signals with high fidelity and great diversity

III. Method

A. Diffusion Probabilistic Models

Diffusion models, also known as diffusion probabilistic models, are a class of latent variable generative models. The fundamental concept involves using a Markov chain to gradually transform one distribution into another through an iterative forward diffusion process. This idea was first applied in non-equilibrium statistical mechanics [34]. Jascha et al. [35] developed a reverse process that can restore structure from a simple known distribution (e.g., a Gaussian distribution). However, it was not until the introduction of the DDPM that diffusion models demonstrated impressive results, particularly in the field of image synthesis.

Following the DDPMs, our diffusion model for generating FHR signals consists of two phases: the forward process and the reverse process, as illustrated in Fig. 1.

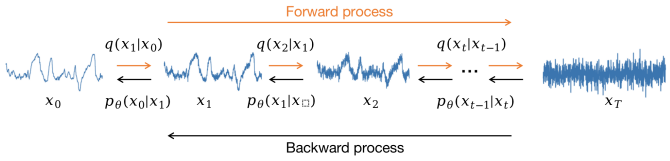


Fig. 1: Forward and Backward Process of the Diffusion Model.

1) Forward Process: Given a data point sampled from a real data distribution $\mathbf{x}_0 \sim q(\mathbf{x})$, the forward process adds random noise to obtain the result of the next step $\mathbf{x}_t$ based on the previous step $\mathbf{x}_{t-1}$. This results in a sequence of noisy samples $\mathbf{x}_1, \ldots, \mathbf{x}_T$, as described in Eq. (1):

$$\mathbf{x}_t = \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\epsilon_t \tag{1}$$

where ϵ_t is the noise at time step t, $\alpha_t := 1 - \beta_t$, and $\beta_1, \ldots, \beta_T$ is the variance schedule that determines the amount of Gaussian noise added to the data.

This process is equivalent to sampling from a Gaussian distribution with mean $\sqrt{\alpha_t}\mathbf{x}_{t-1}$ and variance $(1-\alpha_t)\mathbf{I}$, as shown in Eq. (3):

$$q\left(\mathbf{x}_t \mid \mathbf{x}_{t-1}\right) \sim \mathcal{N}\left(\mathbf{x}_t; \sqrt{\alpha_t}\mathbf{x}_{t-1}, \left(1-\alpha_t\right)\mathbf{I}\right) \tag{2}$$

Furthermore, since $\mathbf{x}_t$ only depends on $\mathbf{x}_{t-1}$ from the previous step, according to the Markov chain property, we can derive the relationship between $\mathbf{x}_t$ and $\mathbf{x}_0$, as in Eq. (3), where $\bar{\alpha}_t = \prod_{i=1}^{t} \alpha_i$:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon \tag{3}$$

2) Reverse Process: The reverse process aims to recreate the true sample from the Gaussian noise input, defined as $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The key idea is to train a model to learn the reverse process $p_\theta\left(\mathbf{x}_{t-1} \mid \mathbf{x}_t\right)$ to reconstruct the original data. The reverse conditional probability is tractable when conditioned on $\mathbf{x}_0$. By leveraging Bayes' rule, we obtain Eq. (4):

$$q\left(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0\right) = q\left(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \mathbf{x}_0\right) \frac{q\left(\mathbf{x}_{t-1} \mid \mathbf{x}_0\right)}{q\left(\mathbf{x}_t \mid \mathbf{x}_0\right)} \tag{4}$$

Since $q\left(\mathbf{x}_t \mid \mathbf{x}_{t-1}, \mathbf{x}_0\right)$, $q\left(\mathbf{x}_{t-1} \mid \mathbf{x}_0\right)$, and $q\left(\mathbf{x}_t \mid \mathbf{x}_0\right)$ can be calculated based on the forward process and are represented by Gaussian distributions, we derive Eq. (5):

$$q\left(\mathbf{x}_{t-1} \mid \mathbf{x}_t, \mathbf{x}_0\right) = \mathcal{N}\left(\mathbf{x}_{t-1}; \tilde{\boldsymbol{\mu}}_t\left(\mathbf{x}_t, \mathbf{x}_0\right), \tilde{\beta}_t \mathbf{I}\right) \tag{5}$$

where $\tilde{\boldsymbol{\mu}}_t\left(\mathbf{x}_t, \mathbf{x}_0\right) := \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t$ and $\tilde{\beta}_t := \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$.

We then define the learnable Gaussian transitions as follows:

$$p_\theta\left(\mathbf{x}_{0:T}\right) = p\left(\mathbf{x}_T\right) \prod_{t=1}^{T} p_\theta\left(\mathbf{x}_{t-1} \mid \mathbf{x}_t\right) \tag{6}$$

and

$$p_\theta\left(\mathbf{x}_{t-1} \mid \mathbf{x}_t\right) = \mathcal{N}\left(\mathbf{x}_{t-1}; \mu_\theta\left(\mathbf{x}_t, t\right), \Sigma_\theta\left(\mathbf{x}_t, t\right)\right) \tag{7}$$

where $\mu_\theta\left(\mathbf{x}_t, t\right)$ and $\Sigma_\theta\left(\mathbf{x}_t, t\right)$ are the mean and covariance matrix of the Gaussian distribution, respectively. These parameters, θ, are learned to control the joint distribution of the reverse process.

The objective is to maximize its variational lower bound (ELBO):

$$\begin{aligned}
&\mathbb{E}_{q_{\text{data}}(x_0)} \log p_\theta(x_0) \\
&= \mathbb{E}_{q_{\text{data}}(x_0)} \log \mathbb{E}_{q(x_1,\cdots,x_T|x_0)} \left[\frac{p_\theta(x_0,\cdots,x_{T-1} \mid x_T) \times p_{\text{latent}}(x_T)}{q(x_1,\cdots,x_T \mid x_0)}\right] \\
&\geq \mathbb{E}_{q(x_0,\cdots,x_T)} \log \left[\frac{p_\theta(x_0,\cdots,x_{T-1} \mid x_T) \times p_{\text{latent}}(x_T)}{q(x_1,\cdots,x_T \mid x_0)}\right] := \text{ELBO}.
\end{aligned} \tag{8}$$

B. PRSA Curve and Spectrogram

Inspired by [33], we use the Phase-Rectified Signal Averaging (PRSA) spectrogram as the label for each signal. The spectral method derived from PRSA better differentiates high-risk pregnancies compared to other traditional spectrum estimators, such as Discrete Fourier Transform (DFT), autoregressive (AR) models, Continuous Wavelet Transform (CWT), and Empirical Mode Decomposition (EMD). The algorithm for generating the PRSA spectrogram is shown in Algorithm 1.

The time duration T for defining deceleration points is set to 1 sample (0.5 s) to make the procedure more sensitive to noise, and the window length L is set to 100 samples. After obtaining the PRSA curve, we compute the scales needed for the wavelet transform. Here, δ_t is the sampling period (set to 0.5 s), s_0 is set to 1 s, and J is equal to 179. A Complex Morlet (CMOR) wavelet is then applied, which is suitable for analyzing oscillatory phenomena in time-series data. Finally, we compute the spectrogram from the wavelet coefficients.

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Algorithm 1 PRSA Spectrogram Computation

Require: FHR series x, window length L, time duration T, FHR series length N, sampling period δ_t,

Ensure: PRSA Spectrogram

1: Define deceleration anchor points (x_{dec}) that satisfy the condition:

$$\sum_{i=0}^{T-1} x[t+i] < \sum_{i=1}^{T} x[t-i] \tag{9}$$

2: **for** each x_{dec} **do**

3: Define a window of length $2L$, which takes the values of the original signal from $x_{\text{dec}} - L$ to $x_{\text{dec}} + L - 1$

4: **end for**

5: Construct the PRSA curve (x_{PRSA}) by averaging all these windows

6: Compute the spectrogram of the PRSA curve using CWT

7: Compute the scales needed for the CMOR wavelet, which is defined as:

$$\begin{cases} \delta_j = J^{-1} \cdot \log_2\left(\frac{N \cdot \delta_t}{s_0}\right) \\ s_j = s_0 \cdot 2^{j\delta_j}, \quad j = 0, 1, \ldots, J \end{cases} \tag{10}$$

8: Compute the wavelet coefficients and spectrogram

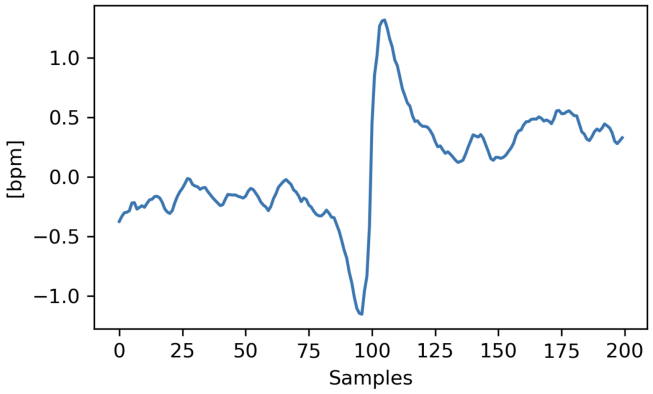


Fig. 2: PRSA curve computed from the FHR series.

The PRSA curve and its spectrogram are shown in Fig. 2 and Fig. 3, respectively.

In Fig. 2, the PRSA curve is used to identify deceleration anchor points in the FHR signal, which are critical for detecting and analyzing fetal health conditions. This curve helps in distinguishing between normal and abnormal fetal heart rate patterns by averaging the signal around specific anchor points.

In Fig. 3, the CWT is applied to the PRSA curve to convert it into a time-frequency representation, providing a detailed view of the frequency components of the signal over time. This spectrogram aids in identifying specific patterns and anomalies in the FHR series by highlighting the variations in signal frequency that correspond to physiological and pathological changes.

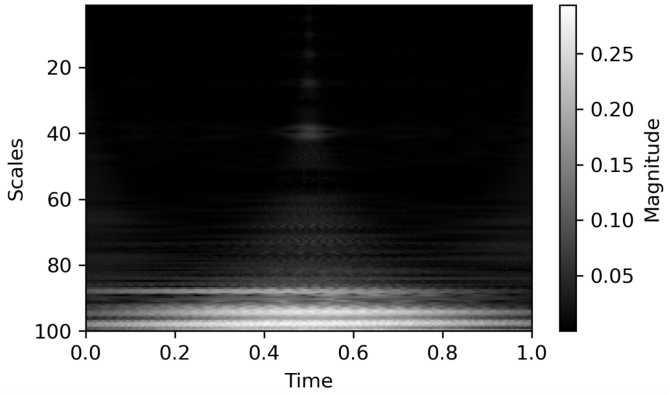


Fig. 3: CWT of PRSA data.

C. Model Architecture

Our model architecture integrates time encoding, condition generation from PRSA spectrograms, and residual blocks with dilated convolutions to effectively manage temporal dynamics and long-range dependencies. The network structure is presented in Fig. 4.

The Time Encoding module converts the diffusion steps into high-dimensional representations using sine and cosine functions. Meanwhile, the PRSA spectrogram generation preprocesses the FHR signals to extract important features, which are then used as conditions for the model. Subsequently, the Condition Generation module upsamples the PRSA spectrograms, preparing them as input conditions for the diffusion process. Additionally, the Residual Blocks, consisting of 30 layers, perform bi-directional dilated convolutions in each layer, thereby incorporating time step and condition information to capture long-range dependencies in the FHR series. Throughout the Forward and Reverse Diffusion Processes, the model progressively adds and removes noise to and from the FHR signals, thereby learning to generate high-fidelity and diverse synthetic signals from Gaussian noise.

1) Time Encoding: To represent the different steps, a 128-dimensional encoding vector is applied [36]. This encoding captures temporal information through a combination of sine and cosine functions. Specifically, three fully connected (FC) layers are used on the diffusion step encoding to extract complex features and map the dimension to the other embedding vectors of the model, facilitating the integration of time-dependent information.

The time embedding vector is defined as follows:

$$\begin{aligned} t_{\text{embedding}} = \Big[&\sin\left(10^{\frac{0\times4}{63}} t\right), \cdots, \sin\left(10^{\frac{63\times4}{63}} t\right), \\ &\cos\left(10^{\frac{0\times4}{63}} t\right), \cdots, \cos\left(10^{\frac{63\times4}{63}} t\right) \Big] \end{aligned} \tag{11}$$

This encoding ensures that the temporal dynamics are effectively captured, enabling the model to better understand and process time-dependent patterns in the data.

2) Condition Generation: As described in Section III-B, each FHR series has a corresponding PRSA curve spectrogram

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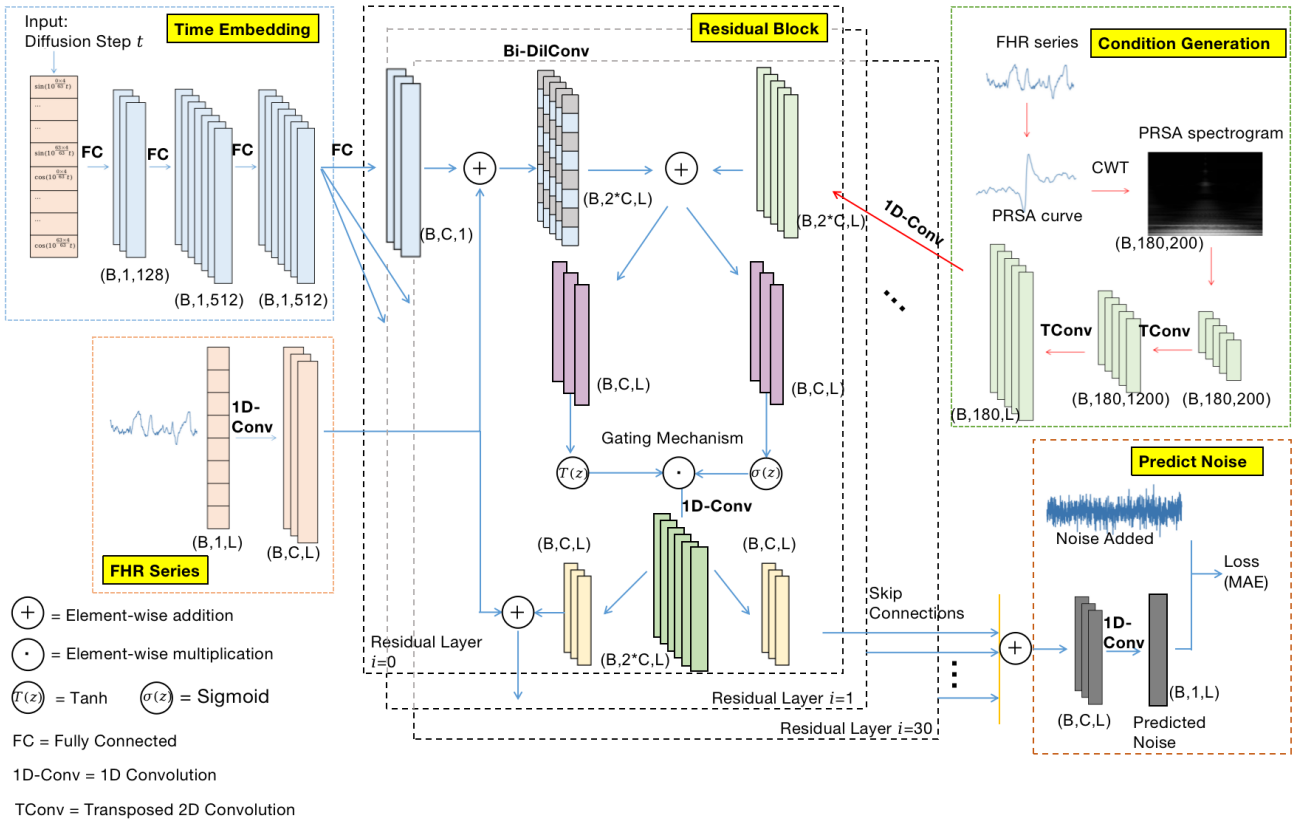


Fig. 4: Network Architecture of FHRDiff for FHR Signal Generation.

that serves as guidance for generation in this model. After pre-processing to extract the PRSA curve and applying the CWT to create the spectrogram, the spectrogram is upsampled through two layers of transposed 2-D convolutions. Subsequently, a 1×1 convolutional layer is applied to map the dimension to 2 residual channels.

3) Residual Block: The model employs 30 residual layers. The output of each layer is split into two separate tensors along the channel dimension: one tensor serves as the input for the next residual layer, while the other tensor acts as a skip connection for the current layer, which will be summed up later. In each residual layer, the diffusion step embedding vector and the FHR series embedding vector are first summed and then passed through a bi-directional dilated convolution. The dilation rate is computed as $2^{i/\text{dilation cycle length}}$, with the dilation cycle length set to 10.

Given the 30 residual layers, this configuration yields a receptive field size of 6,139, which exceeds the input length of 2,400 data points, making it highly effective for capturing long-range dependencies in the FHR series. Additionally, the conditioner, acting as a bias term for the dilated convolution, is incorporated and passed through gated-tanh nonlinearities in each residual layer. This comprehensive design enhances the model's ability to learn complex temporal patterns and dependencies in the FHR data.

IV. Experiment

A. Dataset

This study utilized the CTU-CHB Intrapartum Cardiotocography dataset. The dataset comprises a collection of 552 CTGs from the University Hospital in Brno (UHB) and the Czech Technical University (CTU). Each recording lasts up to 90 minutes and includes a FHR time series, a uterine contraction signal, and detailed maternal, delivery, and fetal clinical information. The signals are sampled at 4 Hz.

B. Pre-processing

A series of signal processing algorithms were applied to the raw signals to eliminate the effects of missing values, baseline wandering, and artifacts. Initially, gaps longer than 60 samples (15 seconds) in the FHR time series were removed. Linear interpolation was then employed for the FHR signal, and signals with more than 10% missing values were excluded from the model inputs. A detrend function was used to eliminate baseline drift. Finally, the long signals were segmented into 20-minute intervals, each containing 4,800 data points. Then the signals were downsampled to 2,400 points. The comparison between raw data and preprocessed data is shown in Fig. 5.

C. Evaluation Results

In line with previous work, we measured and compared the fidelity of the synthetic data using various evaluation metrics.

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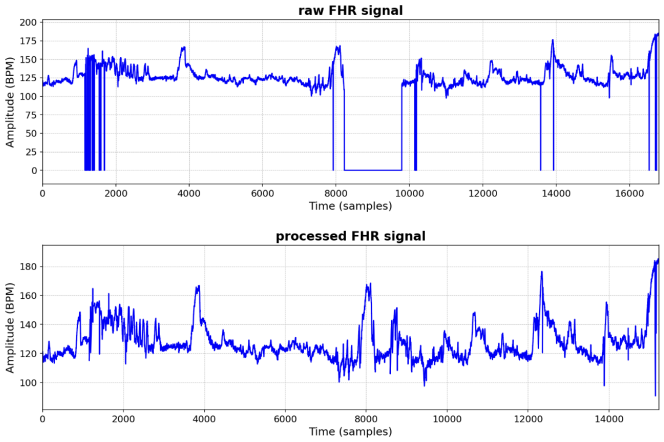


Fig. 5: Comparison of Original Signal and Pre-processed Signal.

To assess the effectiveness of the PRSA curve spectrogram in guiding the model to generate normal and abnormal FHR signals, an ablation study including visual inspection and Principal Component Analysis (PCA) was conducted.

1) Fidelity: High fidelity indicates that the generated data have a distribution similar to the real FHR signals. Several evaluation metrics were applied to determine whether the synthetic data distribution $z \sim P_z$ is indistinguishable from the real data distribution $x \sim P_{\text{data}}$. Relative Entropy (RE), also known as Kullback-Leibler (KL) Divergence, and Frechet Distance (FD) provide different perspectives on whether the generated data capture the characteristics of the real data.

FD measures the similarity between two distributions by finding the minimum possible distance between the two continuous signals. Lower RE and FD values indicate better quality of the synthetic signal. Maximum Mean Discrepancy (MMD), Sliced Wasserstein Distance (SWD), and Percent Root Mean Square Difference (PRD) are also statistical tests used to measure the difference between two different distributions. SWD is computed by averaging the Wasserstein distance over multiple 1-dimensional projections. PRD quantifies the relative difference between two sets of signals.

The evaluation results are shown in Table I. We re-trained five GAN-based architectures to generate FHR signals as baselines: SeqGAN [37], DCGAN [38], CGAN [39], FHRGAN [16], and CTGGAN [27]. Furthermore, we optimized a 2D image-based diffusion model. The one-dimensional (1D) FHR signal was embedded into a two-dimensional (2D) space by mapping the time series signal from Cartesian coordinates to polar coordinates. Gramian Angular Summation/Difference Field (GASF/GADF) and Markov Transition Fields (MTF) were used to construct three separate 2D matrix embeddings of the time series [40]. We then designed a diffusion model with Convolutional Neural Networks (CNNs) as the backbone, incorporating residual connections and an attention mechanism. The details of the original model are referred in [31].

We compared the evaluation metrics among the seven

TABLE I: Comparison of Evaluation Metrics for Different Models

Model	Evaluation Metrics				
	RE (↓)	FD (↓)	MMD (↓)	SWD (↓)	PRD (↓)
SeqGAN [37]	10.324	0.974	6.527	0.541	10.635
DCGAN [38]	9.532	0.795	5.267	0.254	10.431
CGAN [39]	9.012	0.763	5.022	0.254	9.431
FHRGAN [16]	8.059	0.682	1.975	0.086	5.437
CTGGAN [27]	5.621	0.614	0.215	**0.012**	4.821
DDPM [31]	7.059	0.682	6.323	0.521	9.786
FHRDiff	**1.763**	**0.160**	**0.067**	0.088	**0.492**

models. The RE, FD, MMD, and PRD values of our method are significantly lower than those of the others. The evaluation metrics demonstrate that our model surpasses the best GAN by roughly 4× in terms of RE and FD values and 20× in PRD values. These results underscore our model's capacity to generate high-quality FHR data that closely resembles real data.

2) Classification Accuracy: We evaluate the ability of our model to generate synthetic data that closely matches real data, focusing on its capability to generate different FHR signals classified as *normal* or *abnormal*. Given the variability in signal quality in the CTU-CHB dataset—where some signals are recorded during labor and others via ultrasound instead of direct scalp electrode—the signal quality is inconsistent, with some containing significant noise. These factors complicate the use of machine learning for classification tasks.

To address this, we included evaluations of 160 generated samples by an obstetrician in our study. We extracted PRSA curve spectrograms from the test set, which served as conditions for our trained model to generate diverse data. We then calculated the accuracy, precision, specificity, sensitivity, and negative predictive value of the classification.

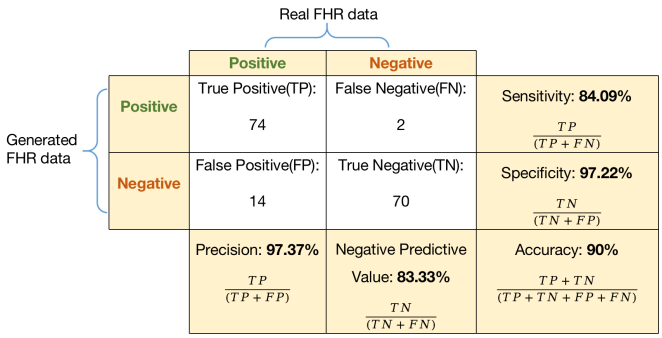


Fig. 6: Evaluation Index of Manually Classification by Obstetrician.

The definitions of the evaluation indices and the results are shown in Fig. 6. The marked values demonstrate that our model can accurately generate different classes of data based on the PRSA curve spectrogram condition.

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D. Ablation Study

Our ablation experiments were designed to explore the hypothesis that conditional diffusion models, when provided with PRSA spectrogram data, produce synthetic FHR signals with greater diversity and closer alignment to physiological norms compared to unconditional models.

1) Visual Inspection: We conducted a visual inspection to assess the diversity of FHR signal samples generated by unconditional and conditional DDPM models. As depicted in Fig. 7, the incorporation of PRSA spectrogram conditions in the DDPM model notably enriches the variability of the synthetic data. It is evident that the data generated by unconditional DDPM exhibit similar patterns at the beginning and end of the signals, while the conditional approach introduces a broader range of patterns.

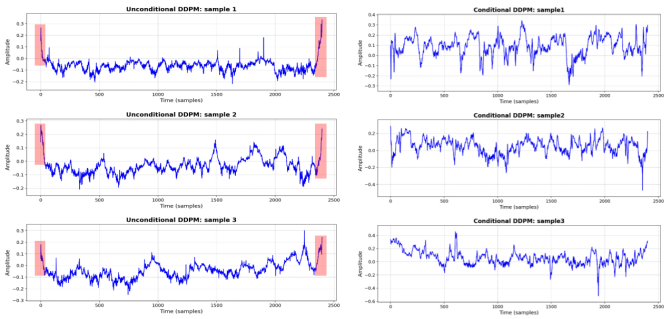


Fig. 7: Visual Comparison of FHR Signal Samples Generated by Unconditional and Conditional DDPM Models.

2) Principal Component Analysis (PCA): To visually assess the impact of PRSA spectrogram conditions on the diversity of generated FHR signals, we employed Principal Component Analysis (PCA). Given the inherently one-dimensional nature of FHR signals, we transformed them into two-dimensional representations using Summation of GAF (GASF). The conversion process is as follows:

1) Normalize each FHR signal to the range $[-1, 1]$, denoted as x_i.
2) Determine the corresponding angles using the formula:

$$\theta_i = \arccos(x_i), \quad x_i \in [-1, 1] \tag{12}$$

3) Generate the GASF matrix with:

$$\text{GASF}_{ij} = \cos(\theta_i + \theta_j) \tag{13}$$

where θ_i and θ_j represent the angles for the i-th and j-th signal points, respectively.

We randomly generated 30 datasets for both conditional and unconditional models. After converting these datasets into GASF representations, PCA was conducted to reveal the underlying patterns. The outcomes are depicted in Fig. 8, illustrating the dispersion and diversity of the generated data.

The PCA plot indicates that samples from the unconditional model are closely clustered, suggesting limited diversity. In contrast, samples from the conditional model are more dispersed, indicating a broader range of signal patterns. This observation underscores the enhanced diversity and variability in FHR signals generated under the conditional model, which aligns with our expectations for physiological signal representation.

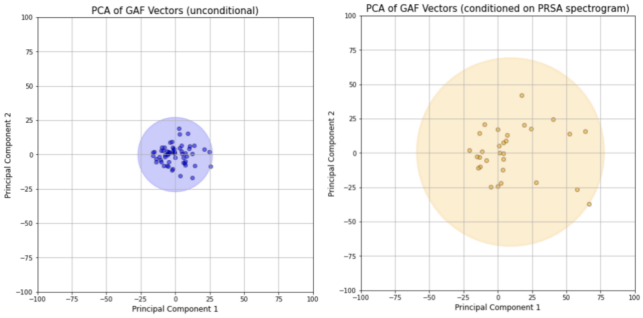


Fig. 8: Comparison between Unconditional and Conditional models with PRSA spectrogram using PCA.

V. Discussion and Limitation

Our study demonstrates the exceptional performance of the proposed FHRDiff model in generating high-fidelity FHR signals across various categories. Despite these promising results, it is imperative to recognize and address the limitations inherent in our research.

First, the expert evaluation was conducted by a single clinician, which could introduce subjectivity and bias into the assessment. To mitigate this, future studies should engage a panel of clinicians, which will not only diversify the expertise but also enhance the reliability and generalizability of the evaluation. Second, the determination of an adequate sample size for a statistically robust evaluation is a critical consideration that was not fully addressed in our study. The aforementioned two limitations may result in lower inter-rater reliability, and this sample size may not be sufficiently powered to detect the difference between the diagnosis from different clinicians. Future research should incorporate a statistically justified sample size to ensure the validity and power of the findings. Moreover, while expert assessments provide qualitative insights, they are limited in scope. Integrating machine learning techniques to quantitatively analyze the performance of the model on datasets with varying ratios of real to synthetic data could offer a better understanding of the model's capabilities and limitations.

By addressing these limitations, subsequent research can build upon the foundation laid by our study, refining the model's application and expanding its utility in the field of fetal health monitoring.

VI. Conclusion

This study introduced a novel diffusion probabilistic model conditioned on PRSA spectrograms for synthesizing FHR signals. Our model captures essential physiological and pathological characteristics, generating high-fidelity and diverse data.

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Utilizing the CTU-CHB dataset, we demonstrated superior performance over GAN-based models and a 2D diffusion model across multiple metrics. Expert validation confirms the clinical realism of our synthetic signals.

The model's success in mimicking FHR signals suggests significant potential for improving fetal health monitoring and diagnostic tools. Future work will refine the model and extend its application to other biomedical time-series data. In summary, our conditioned diffusion model offers a promising approach for synthetic FHR signal generation, with valuable applications in research and clinical practice.

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